

AML Homework 2 - Linear regression

Due on February 5, 2026 at 11:59 pm

In this homework, you will experiment with linear regression, using Python in combination with in lecture discussed ML, data processing, and data visualization packages. For this purpose, write a program that performs the analysis steps as described below.

- a) First, load the “*Auto.csv*” data set, and use the feature “*displacement*” (input) as a predictor for “*mpg*” of the car (output).
- b) For training and testing of the model, split the dataset randomly into a training set (60%) and test set (40%).
- c) Fit the linear regression model to the training dataset. Report the slope and intercept of the model.
- d) Evaluate model performance utilizing the test dataset. For this purpose, calculate and report several suitable performance metrics.
- e) Generate a plot showing test data and the trained model (regression line).
- f) Repeat the steps c) to e) for the following features as input: “*cylinders*”, “*weight*”, and “*acceleration*”.
- g) Rank all four features regarding their ability to predict the “*mpg*” of the car (i.e., from best to worst).

Document your work, results (numbers, plots, etc.), your analysis of results (e.g., did everything work as you expected, etc.), and conclusion in a report. Submit your Python code and report (PDF format) on ICON.